



Microbiome: An Emerging New Frontier in Graft-Versus-Host Disease

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Abstract

Hematopoietic cell transplantation is an intensive therapy used to treat high-risk hematological malignant disorders and other life-threatening hematological and genetic diseases. Graft-versus-host disease (GVHD) presents a barrier to its wider application. A conditioning regimen and medications given to patients undergoing allogeneic hematopoietic stem cell transplantation (allo-HCT) are capable of disturbing the homeostatic crosstalk between the microbiome and the host immune system and of leading to dysbiosis. Intestinal inflammation in the context of GVHD is associated with loss in microbial diversity that could serve as an independent predictor of mortality. Successful gastrointestinal decontamination using high doses of non-absorbable antibiotics likely affect allo-HCT outcomes leading to significantly less acute GVHD (aGVHD). Butyrate-producing Clostridia directly result in the increased presence of regulatory T cells in the gut, which are protective in GVHD development. Beyond the microbiome, Candida, a member of the mycobiome, colonization in the gut has been considered as a risk factor in pathophysiology of aGVHD and reduction in GVHD is observed with antifungal prophylaxis with fluconazole. Reduced number of goblet cells and Paneth cells have been shown to associate with GVHD and has a significant impact on the micro- and mycobiome density and their composition. Lower levels of 3-indoxyl sulfate at initial stages after allo-HCT are related with worse GVHD outcomes and increased mortality. Increased understanding of the vital role of the gut microbiome in GVHD can give directions to move the field towards the development of improved innovative approaches for preventing or treating GVHD following allo-HCT.

Keywords Microbiome · Hematopoietic cell transplantation · Graft-versus-host disease · Dysbiosis

Introduction

The mammalian gastrointestinal (GI) tract is mainly colonized with a highly diverse complex and dynamic microbial community consisting of bacteria, but also contains fungi, archaea, viruses and eukaryotes with a density of approximately 10^{13} – 10^{14} cells/g fecal matter. It is dominated by

anaerobic Firmicutes (~ 51%) and Bacteroidetes (~ 48%). The remaining 1% consists of Proteobacteria, Verrucomicrobia, Fusobacteria, Cyanobacteria, Actinobacteria and Spirochetes, including various species of fungi, protozoa, viruses and other microorganisms [1, 2]. Commensal microbes have tremendous potential to impact our physiology, both in health and in disease via contributing in metabolic functions, protecting against pathogens, and educating the immune system. A gut microbiota at symbiosis contains a balanced composition of symbionts and pathobionts. Pathobionts are potentially pathological organism which lives as a symbiont under normal circumstances [3]. However, when the homeostasis is disturbed, dysbiosis has the potential to induce inflammatory stimuli and to cause imbalance of T-helper 17 (Th17) and regulatory T cells [4]. An example how changes in the microbiome can affect our daily live is reflected in numerous studies supporting the “hygiene hypothesis”, suggesting that a lack of early childhood exposure to diverse microorganism increases susceptibility to allergic diseases [5].

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Analysis of the gut microbiome has been largely dependent on *ex vivo* cultivation of bacteria. These techniques are only able to cultivate 10–30% of gut microbiota, leading to a large knowledge-gap in gut microbiome composition. In recent years, the developments of techniques based on genetic analysis have facilitated the extensive exploration of the gut microbial ecosystem. These recently developed next-generation sequencing technologies, such as 16S rDNA sequencing and metagenomics, have allowed for further identification and an increased depth of analysis of microorganisms that are otherwise unable to be cultured. This has helped reveal the bacterial composition of different anatomic environments, including the human gut [2].

Hematopoietic cell transplantation (HCT) is an intensive therapy used to treat high-risk hematological malignant disorders and other life-threatening hematological and genetic diseases. Graft-versus-host disease (GVHD) presents a barrier to its wider application [6], as it is the major cause of morbidity and mortality after allogeneic HCT (allo-HCT). GVHD is an immunologically mediated process, characterized by donor T cell responses to host allo antigens and the dysregulation of inflammatory cytokines after allo-HCT [7]. The immunological attack on target recipient organs and tissues by donor allogeneic T cells causes tissue damage in the GI tract, liver, lung, and skin. Recipient age, toxicity of the conditioning regimen [like total body irradiation (TBI), antibiotics use], hematopoietic graft source, and acute GVHD (aGVHD) prophylaxis approaches are factors which affect the development and severity of GVHD.

GI GVHD is the major cause of mortality [8]. The microbiota and their metabolites help build the immune system and maintain intestinal homeostasis, and they may alter host susceptibility to GVHD. Various studies have shown that during allo-HCT, the intestinal mucosa is damaged, and that the commensal microbiome is affected, leading to a dysbiotic state with reduced diversity. Several important outcomes after allo-HCT, in particular, bloodstream infections, GVHD, organ toxicity, and overall survival, have been associated with intestinal bacterial composition [9–13]. Advanced technologies have enabled researchers to study the detailed and specific role of intestinal microbiota in GVHD pathophysiology.

Gut Microbiome and GVHD

Patients undergoing allo-HCT are exposed to numerous factors, such as chemotherapy, radiation, and antibiotics, which together and individually can disturb the homeostatic cross-talk between the microbiome host immune systems. The impact of the microbiota on GVHD is known to be significant and complex [14–16].

Impact of Bacterial Diversity on GVHD and Role of Specific Microbes

Intestinal inflammation in the context of GVHD is associated with major shifts in the composition of the intestinal microbiota in murine as well as human recipients of allo-HCT. Large shifts within the phylum Firmicutes were observed, with a dramatic increase in Lactobacillales and decreases in Clostridiales in mouse models of GVHD. Interestingly, though, in the same study, elimination of Lactobacillales before HCT aggravated GVHD in mice, and reintroduction of the predominant species of *Lactobacillus* resulted in significant protection against GVHD. Similar microbiome patterns in patients were seen during onset of intestinal inflammation caused by GVHD [14]. Loss of diversity in intestinal microbiota with simultaneous expansion of *Lactobacillus* species and contracture of the commensal bacterial flora has been observed in both mice and patients with GVHD by others as well [14, 15]. Another study reported protection from experimental GVHD by the prophylactic application of probiotic lactobacilli in mice [16].

Recently it has been shown that maintaining a diverse population of symbiotic and mutualistic bacteria decreases inflammatory bowel disease [17] and HCT [9, 14, 18]. It has been demonstrated that the intestinal microbiota diversity at engraftment served as an independent predictor of mortality in allo-HCT recipients and it was observed that mortality outcomes were significantly worse in patients with lower intestinal diversity. Allo-HCT recipients showed overall survival at 3 years of 36%, 60%, and 67% for low-, intermediate-, and high-diversity groups, respectively [9]. Other study also reports decreased bacterial diversity in GVHD patients [14, 15, 18].

Bacteremia is one of the complications of allo-HCT as a result of transplant-related conditioning due to disrupted mucosal barrier function [19]. Vancomycin-resistant *Enterococcus* (VRE), viridians-group *Streptococcus*, and aerobic gram-negative bacteria are considered the most common causes of bloodstream infection post allo-HCT [20]. It is suggested that indomination by bacteria during allo-HCT as a result of disrupted diversity and stability of the intestinal flora are cause for subsequent bacteremia. Patients with enterococcal domination showed a ninefold increased risk of VRE bacteremia, and domination by *Proteobacteria* increased the risk of bacteremia with aerobic gram-negative bacilli fivefold [21]. Antibiotic treatment is important in transplant-related shift in the microbiome [22]. Assessment of fecal microbiota during allo-HCT may help to identify patients at greater risk for bacteremia and assist in modifying the treatment for allo-HCT recipients.

Two independent cohort analyses from the same study showed reduced GVHD lethality in patients harboring

increased amounts of bacteria belonging to the genus *Blautia*. Higher abundance also showed correlation with improved overall survival. Clinical factors like treatment with antibiotics that inhibit anaerobic bacteria and receiving total parenteral nutrition for longer durations affected the abundance of *Blautia* causing a decrease. The increased abundance of commensal bacteria belonging to the *Blautia* genus is associated with reduced lethality of GVHD and improved overall survival [15].

Microbes and Graft-Versus-Leukemia Effects

Relapse of primary malignancy remains the leading cause of death after allo-HCT [23]. Studies have reported that the bacterial microbiome plays an important role in normal hematopoiesis [24–26].

In a study by Peled et al., it was reported that higher abundance of a bacterial group composed mostly of *Eubacterium limosum* was associated with a decreased risk of relapse/progression of leukemic disease. Incidences of relapse/progression among patients were 19.8% and 33.8% with and without this group of bacteria, respectively. This association was strongest among recipients of T-cell—replete allografts. They suggested that the *E. limosum* group might serve as potential biomarkers or therapeutic targets to prevent relapse and improve survival after allo-HCT [12]. Findings revealing the association of a particular group or species of bacteria with allo-HCT outcomes are of vital importance, as targeting these bacterial groups may improve survival after allo-HCT. Further evidence for a role of the intestinal microbiome in GVHD was suggested by a study where T cells harvested from germ-free (GF) donors and specific pathogen-free (SPF) donors did not affect GVHD progression, severity and survival and histopathological scores of the small intestine and liver [16, 27].

The prolonged antibiotic use impacting the microbiome composition are associated with adverse hematologic effects, including anemia and neutropenia [28], and specific components of the microbiome which influence normal blood production need to be considered for development of novel treatment modalities.

Virome and GVHD

Viral infections are a major challenge in allogeneic transplant, as they pose both a direct threat to the patient and they can interfere indirectly with patient outcome due to their effects on GVHD. Cytomegalovirus (CMV) reactivation after HCT has been associated with increased morbidity and mortality, and a causal relation between GVHD and immunosuppressive therapy and vice versa has been postulated

[29]. Epstein–Barr virus (EBV), another member of the herpes virus group, rarely associates with mortality [30], yet can give rise to post-transplantation lymphoproliferative disorders (PTLDs) with possible fatal outcome. Further, BK and John Cunningham viruses can reactivate post-bone marrow transplant (BMT) [31]. The bacterial microbiota influence immunity to a variety of viruses. Recently, it has been shown that preexisting antibodies to enteric bacteria can skew vaccine responses to cross-reacting HIV-1 antigens, arguably rendering a vaccine less protective [32]. Enteric bacterial components regulate vaccine responses to influenza in mice through activation of the innate immune receptor, Toll-like receptor 5 (TLR5) [33].

The microbial interactions are not as simple as they look. It is a complex interplay between microbiome, mycobiome and virome which needs to be explored at a large scale. Recent reports suggest that enteric viruses regulate and are regulated by these other microbes through a series of processes termed “transkingdom interactions” [34]. Intestinal bacteria are able to promote replication and transmission of enteric viruses from four different families: noroviruses, picornaviruses, retroviruses and reoviruses [35–38]. Norovirus infection can cause severe disease with significant morbidity and mortality after allo-HCT, both before and after engraftment [39]. Picornaviruses commonly known to cause “the common cold” in the general population may cause severe infections in immunocompromised hosts such as HCT recipients [40].

Mycobiome and GVHD

Although gut bacteria are currently the major focus in GVHD research, the importance of fungi cannot be underestimated. *Candida* species colonization (a member of the mycobiome) in the gut has been considered as a risk factor in pathophysiology of aGVHD, and reduction in GVHD is observed with antifungal prophylaxis with fluconazole. Dectin-1 is a pivotal pattern-recognition receptor that recognizes the β -D-glucan found in fungal cell walls, including those of *Candida* and triggers mucosal antifungal CD4 + Th17 responses. The release of IL-17 and IL-22 is a key aspect of mucosal antifungal defenses. C-type lectin receptor recognizes the α -mannan (another class of polysaccharides) cell wall component of *Candida* species. This stimulates macrophages to produce cytokines, such as IL-6 and IL-23, which drive a Th17-skewed response on recognition by the C-type lectin receptor [8, 41]. These fungal wall components are suspected in GVHD pathophysiology. C-type lectin-mediated Th17 responses have recently been implicated in the pathogenesis of aGVHD. Th17 responses have been shown to play an important role in both mucosal and systemic anti-*Candida* host defenses in mice and humans [42].

These responses are induced through the activation of different pattern recognition receptors (PRRs), most importantly dectin-1, dectin-2, mannose receptor, and TLR-2. Defective innate immune responses caused by dectin-1 deficiency results in increased colonization with *Candida* spp. and a decreased Th17 response [42].

Molds can invade through a breach in the skin or in the GI tract integrity. This is seen during periods of GI tract mucositis secondary to chemotherapy toxicity or GVHD, allowing further dissemination of luminal content of GI tract [43, 44]. Mold infections, especially those caused by *Aspergillus fumigatus*, were recognized as occurring at greater incidence (5–12%) in allogeneic HCT during the 1990s [45, 46] and associated with 80–90% mortality [47]. In addition to *Aspergillus fumigatus*, other aspergillus species including *A. terreus* and *A. flavus* are able to cause invasive disease, just as other molds like *Fusarium* species, *Scedosporium* species, and *Zygomycetes* [48]. However, many questions regarding fungal infections and their composition are not very well answered and require further investigation of mycobiome profiling in allo-HCT patients and correlation with disease outcome.

Antibiotic Prophylaxis and Treatment and Microbes

Elimination of facultative and strict anaerobic microorganisms from the GI tract by antimicrobial drugs in the period of time around allo-HCT potentially prevents aGVHD. Use of Rifaximin for decontamination on intestinal microbiota composition seemed useful over ciprofloxacin/metronidazole. Rifaximin use helped to maintain high intestinal microbiome diversity and diminished the negative effect of systemic antibiotics on microbial composition. Lower transplant-related mortality at 1 year and overall higher survival were seen with Rifaximin use [49]. Patients on Rifaximin showed lower enterococcal positivity and higher urinary 3-indoxyl sulfate concentrations after HCT [49], suggesting a role for the enterococcus bacterial group. A member of the Enterococcus group, *Enterococcus faecalis* has been shown to affect the mucosal barrier and to be involved in intestinal inflammation through its metalloprotease [50]. Therefore, selection of appropriate antibiotics targeting specific bacteria like enterococcus may influence the gut microbiome and the overall survival following allo-HCT [51]. It has further been shown that microbial diversity is inversely correlated with plasma protein biomarkers ST2, TNFR1 and REG3 α . These biomarkers have emerged as noninvasive tools to diagnose and prognosticate aGVHD [22], and therefore, the relation between GVHD biomarkers and intestinal microbiota might serve as a base for designing future studies for therapeutic interventions.

Total decontamination of the gut using high doses of non-absorbable antibiotics alters the allo-HCT outcomes, and significantly less aGVHD was observed with successful total GI decontamination (GID) [52], suggesting that the translocation of luminal bacteria and their wall compounds might be inhibited during total GID, hereby reducing PRR signaling and subsequent inflammation and immune activation.

Allo-HCT recipients are exposed to an increased risk for clostridium difficile infections (CDIs) secondary to their exposure to chemotherapy, radiation, and antibiotics. It has been demonstrated that approximately 15–30% of allo-HCT recipients develop CDI during transplantation [53, 54]. Recurrent CDI is difficult to treat as antibiotic therapy often fails. Fecal microbiota transplantation (FMT) has been proved as an efficacious method for the treatment of CDI [53, 55]. It has been shown that duodenal infusion of donor feces in patients with recurrent CDI resulted in significant resolution with no significant side effects [56]. Furthermore, in addition to resolution of CDI, fecal infusion resulted in increased bacterial diversity and an increase in Bacteroidetes species and clostridium clusters IV and XIVa and a decrease in Proteobacteria species [56]. However, not all clostridia strains are considered to be harmful.

It was recently hypothesized that reversal of intestinal dysbiosis could improve intestinal GVHD. In a pilot study, the safety and effectiveness of fecal microbiota transplantation (FMT) as treatment for steroid-dependent or steroid-refractory intestinal GVHD has been assessed. Single FMT appeared safe and is effective in allogeneic HCT recipients and no serious adverse events were observed that could be attributed to the FMT. However, one patient showed rapidly progressive hyperacute GVHD, the course of which did not change after FMT, and died from multi-organ failure 7 days after FMT [57].

Microbial Metabolites and Their Contribution to Epithelial Integrity and Immunoregulation

The metabolites that derive from microbial transformation of dietary components have significant effects on physiological processes such as gut and immune homeostasis, energy metabolism, vascular function, and neurological behavior. The alterations of GI microbiota-derived short-chain fatty acids (SCFAs) are well known in various diseases [58, 59]. Butyrate, one of the SCFAs, is known to modulate histone acetylation [60], also serves as an energy source for intestinal epithelial cells (IECs) [61] and is diminished in the intestinal tissue after allo-HCT; however, other SCFAs remained unaffected [62]. Authors have observed that increasing butyrate via intra-gastric gavage restored acetylation of histone H4 in IECs and mitigated GVHD with concomitant decrease in

apoptosis, improved IEC junctional integrity, and decreased apoptosis [63, 64]. Another study reported that administration with 17 rationally selected strains of high-butyrate-producing Clostridia directly results in the increased presence of regulatory T cells (Tregs) in the gut, which play a critical role in maintaining GI homeostasis by modulating inflammatory responses via the release of the anti-inflammatory molecule IL-10, which also directly impact macrophages [64, 65]. The administration with 17 rationally selected strains of high-butyrate-producing Clostridia also resulted in decreased GVHD [63, 66]. In addition, regulatory T cells have been found to be protective in GVHD development [67], likely via their suppressive effects on alloreactive T cell expansion and T effector cell proliferation.

Indole is produced from L-tryptophan by commensal bacteria-expressing tryptophanase and is the origin of the smell of normal human feces. These normal-smelling feces may be an underestimated health indicator [68]. Indole functions as a ‘quorum-sensing’ signal that regulates the virulence and biofilm formation and also serves as modulator of mucosal barrier function and expression of pro- and anti-inflammatory genes by IECs [18, 68]. It has been shown that low 3-indoxyl sulfate (3-IS) levels in urine specimens of allo-HCT recipients at an early stage (first 10 days after HCT) were associated with significantly higher transplant-related mortality at 1 year after allo-HCT [69]. Similarly, in other study conducted by Holler et al. [18], stool specimens from patients with active GVHD had reduced indoxyl sulfate levels from 42.5 ± 11 mmol/L to 11.8 ± 2.8 mmol/L in all post-transplant samples and to 3.5 ± 3 mmol/L. Early reduction of 3-IS levels were seen to be affected by type of GI decontamination, early onset of antibiotic treatment, and recipient NOD2/CARD15 genotype. The reduction in 3-IS levels in urine specimens was correlated with the loss of bacterial diversity. High urinary 3-IS levels were associated with the members of the families of Lachnospiraceae and Ruminococcaceae of the class of Clostridia and low 3-IS levels were found to be associated with the members of the class of Bacilli [69]. The reduction in 3-IS levels in stool specimens was correlated with a relative shift from a predominant commensal bacteria at the time of admission toward enterococci. This shift was pronounced in patients that developed active GI GVHD and was more prominent under antibiotic prophylaxis and treatment of neutropenic infections. Furthermore, predominance of *Enterococcus faecium* or both *E. faecium* and *Enterococcus faecalis* were discovered in these specimens through strain-specific enterococcal polymerase chain reaction (PCR) analysis. The author also mentioned that it is important to note that inclusion in the model does not necessarily mean that these operational taxonomic units (OTUs) are either capable or incapable of converting tryptophan to indole. OTU is an operational definition used to classify groups of closely related individuals.

The model may also include OTUs that do not produce indole themselves but grow either better or worse in the presence of indole [18]. This demonstrates the relevance of microbiota-derived indole and hereby of its importance in mucosal integrity and protection from inflammation.

Impact of Host Physiology on the Microbiome and GVHD

Genetic predisposition has been reported to be capable of influencing the gut microbiome and hereby impacting the transplant outcome. The fucosyltransferase 2 (FUT2) gene produces an enzyme that is responsible for the synthesis of the H antigen in body fluids and on the intestinal mucosa. FUT2 genotype influences risk of acute GVHD and bacteremia after HCT through modifying the microbial composition. This might involve altered intestinal surface glycosylation. The nonsecretory AA genotype of FUT2 is associated with a reduced risk of bacteremia [70]. Reduced bacterial diversity, richness, and depletion of Firmicutes abundance and Bifidobacteria with subsequent increased representation of *Blautia*, *Clostridium sphenoides*, *Ruminococcus gnavus*, and *Doreformicigenerans* in nonsecretor genotypes has been reported [71, 72].

Gut Epithelium Damage During GVHD

Goblet cells secrete mucin and create a mucosal barrier from luminal contents [73]. Penack et al. [74] have shown that Goblet cell injury in GVHD leads to translocation and dissemination of dominant luminal pathogens, which further accelerates GVHD and infections. Paneth cells are important regulators of microbial density in the small intestine. These cells secrete antimicrobial peptides, including α -defensins and Reg-III γ [75]. Lower numbers of Paneth cells have been found during GI GVHD and Paneth cell number inversely correlated with clinical GVHD severity and outcome [76, 77]. Paneth cell number count seems to be correlated with response to GVHD therapy and thus can be used as marker for grading of GI GVHD histopathology [77]. Inflammatory conditions, such as inflammatory bowel disease, show reduced numbers of Paneth cells and changes in the intestinal microbiota. This suggests that maintaining or inducing the Paneth cells to secrete sufficient amounts of antimicrobial peptides controlling the microbial imbalance might be helpful in regulating GVHD development. How could this be the case? A healthy crosstalk is established between the host immune system and the gut microbiota during homeostasis, contributing in metabolic functions, protecting against pathogens, and educating the immune system. IECs possess a range of PRRs that stimulate production and release

abundant antimicrobial peptides, including α -defensins, which are synthesized and stored in Paneth cell granules without the need for sensing of microbe-associated molecular patterns (MAMPs). However, other antimicrobial peptides, like Reg-III γ , require signals from the microbiota for its synthesis. These factors have been shown to control the composition of the microbiota [75]. During normal conditions, the systemic translocation of bacteria or bacterial products through the intestinal mucosal barrier is inhibited by controlling the excessive interaction of MAMPs with host immune cell inflammatory reaction [78]. However, when the homeostasis is disturbed (dysbiosis), translocation and dissemination of luminal content begins.

Several commensal bacteria or their components have the ability to protect intestinal mucosal tissue from irradiation damage [79]. Paneth cells and goblet cells play a key role in limiting bacterial penetration into host tissues and are critical to maintenance of homeostatic crosstalk

between gut flora and the immune response in the small intestine [73, 74, 77]. On disruption of the gut homeostasis, translocation of bacteria and their antigens and other food antigens induces innate immune cells of the host that express certain PRRs which detect pathogen-associated molecular patterns (PAMPs) associated with microbes. This recognition of PAMPs through PRRs results in the activation of various immune cells, such as neutrophils and antigen-presenting cells (APCs), results in APC activation and increased T cell stimulatory capacity. In addition to these cells of hematopoietic origin, epithelial and endothelial cells also express PRRs [11], resulting in increased major histocompatibility complex (MHC) expression and inflammatory cytokine production, together propelling inflammation of the allogeneic T cell response with all of its consequences of target organ infiltration, apoptosis induction and clinical GVHD. Figure 1 depicts the overview of microbiome impact in GVHD pathophysiology.

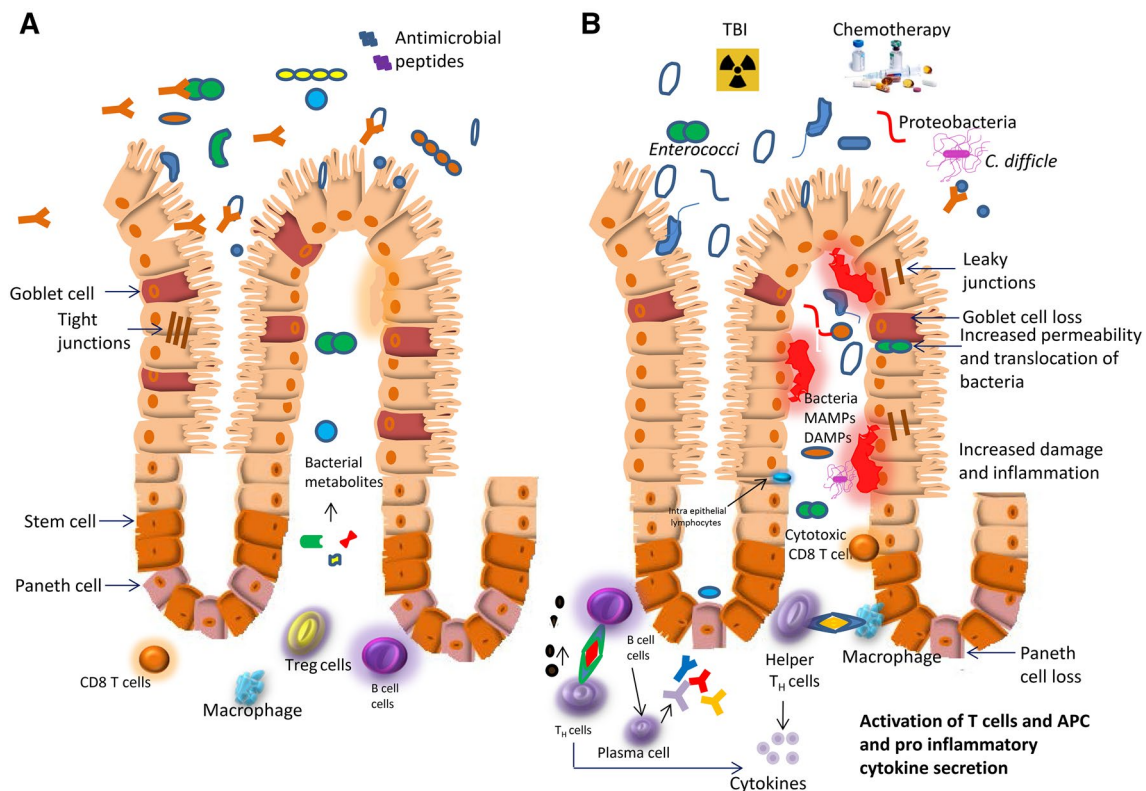


Fig. 1 Schematic overview of microbes and GVHD physiology: **a** During the homeostatic condition gut epithelium maintains its integrity and translocation of luminal content is under regulation. Paneth cells secrete antimicrobial peptides which regulate the pathobionts growth and mucus secreted by goblet cells inhibit bacterial translocation. **b** After conditioning therapy the epithelial barrier is injured leading to leaky junctions and the microbiome niche gets disturbed in its homeostatic condition, reducing beneficial bacteria and over-

growth of harmful bacteria. Leaky junctions allow translocation of bacteria, MAMPs (microbe-associated molecular patterns), and DAMPs (danger associated molecular patterns) to the mucosa. These are recognized by toll like receptors and presented by host antigen-presenting cells (APCs) to T cells leading to the cytokine storm phase and promotes T cell activation and proliferation, resulting in the effector phase causing organ damage and aggravating GVHD

Discussion

The advancements in technologies and tools have facilitated discoveries and defined a crucial role of the microbiome associated with other organ systems beyond the huge gut microbiome itself. Various studies have revealed major microbiome shifts and a loss of diversity of the intestinal microbiome in the course of allo-HCT that mainly occur in the period of antibiotic treatment that affects the intestinal inflammation in the setting of allo-HCT. Studies suggest that the diversity of the intestinal microbiota at engraftment is an independent predictor of mortality in allo-HCT recipients. It is indicated that disrupted diversity and stability of the intestinal flora might result in domination by bacteria associated with subsequent bacteremia.

Although specific components of the microbiota that might be a possible risk factor for aggravating or ameliorating aGVHD are not well understood, research has indicated the preventive role of Lactobacillales, Clostridiales, the *Eubacterium limosum* group and genus *Blautia*, Bacteroidetes species and clostridium clusters IV and XIVa in GVHD as they were found to be associated with less GVHD. These reports suggest that restoration of these strains in GVHD patients might be helpful. In contrast, higher Enterococcal and Proteobacteria species and *C. difficile* abundance have been suspected in aggravating GVHD. However, studies representing the translational application of these observations are lacking and require extensive analysis.

Recent studies on experimental models of GVHD suggests that alteration of the microbiome and subsequent aggravated GVHD could be due to damaged intestinal mucosa, mainly related to Paneth and goblet cell loss. Thus, approaches to allow improved survival or restoration of the epithelium may prove helpful regulating overgrowth of pathobionts and dissemination of MAMPs, PAMPs and damage-associated molecular patterns (DAMPs). Microbial metabolites like butyrate, an SCFA, and indole are indicators of how the microbiome associates and influences GVHD. They are capable of modulating the mucosal barrier function and the expression of pro- and anti-inflammatory genes. However, studies using these microbial metabolites as a therapeutic approach are still not performed extensively. The FMT method used to resolve CDI has emerged as an efficacious method for the treatment of severe CDI. Growing understanding of the vital role of gut microbiome in GVHD will help to propel the field forward and provide the basis in designing more studies towards the development of improved innovative approaches for preventing and treating GVHD following transplantation, including possibly FMT of specific bacteria-enriched products. In addition to exploring the

bacterial microbial changes during GVHD, focus needs to be directed towards mycobiome and virome composition. As the host microbiome consists of different kingdoms, studying interactions among transkingdoms is of great importance yet challenging. Studies focusing on the functional genomics of host microbiomes may provide a better picture of complex transkingdom interaction and may prove helpful in designing therapeutic approaches.

Compliance with ethical standards

Conflict of interest There is no conflict of interest.

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